

A head of broccoli made out of modeling clay, smiling in the sun



Figure: An example of text-to-image generation by DALL.E 2

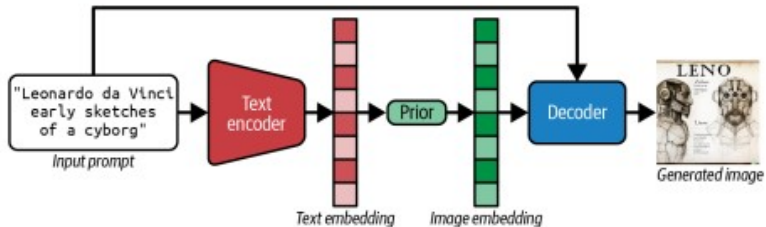


Figure: The DALL.E 2 architecture

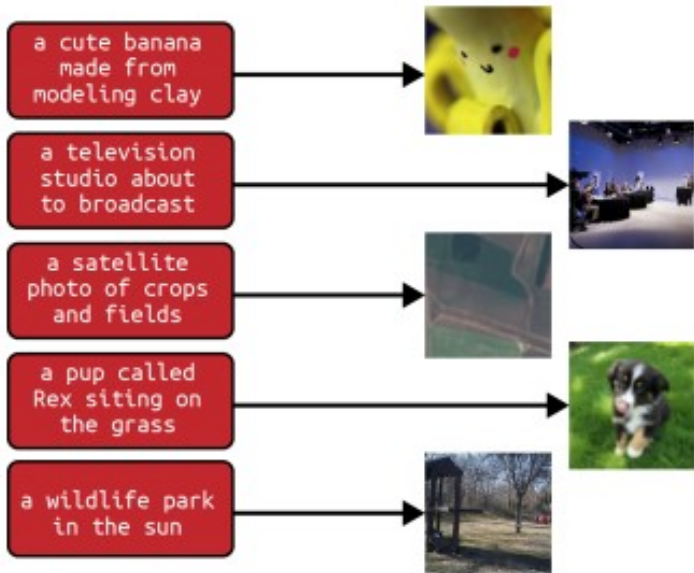


Figure: Examples of text-image pairs

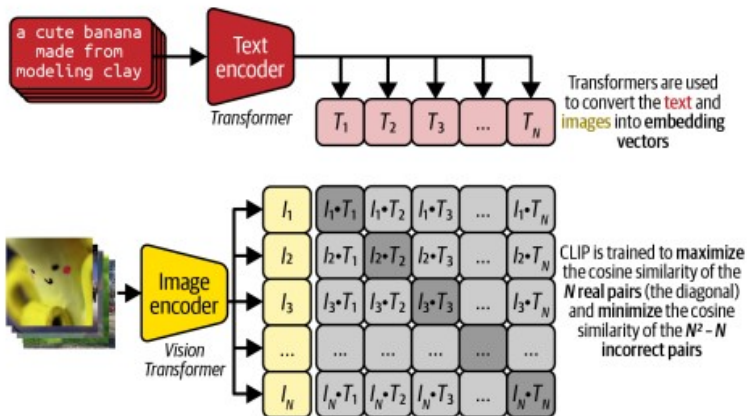


Figure: The CLIP training process

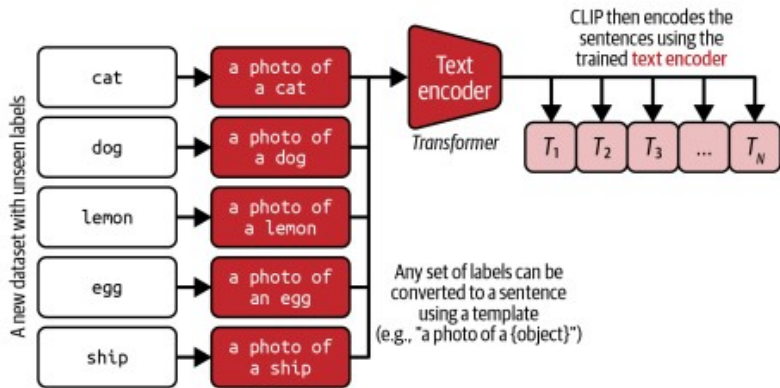


Figure: Converting labels in a new dataset to captions, in order to produce CLIP text embeddings

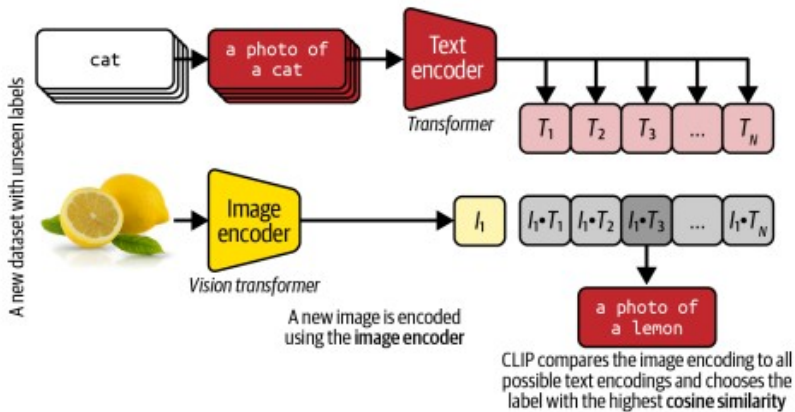


Figure: Using CLIP to predict the content of an image

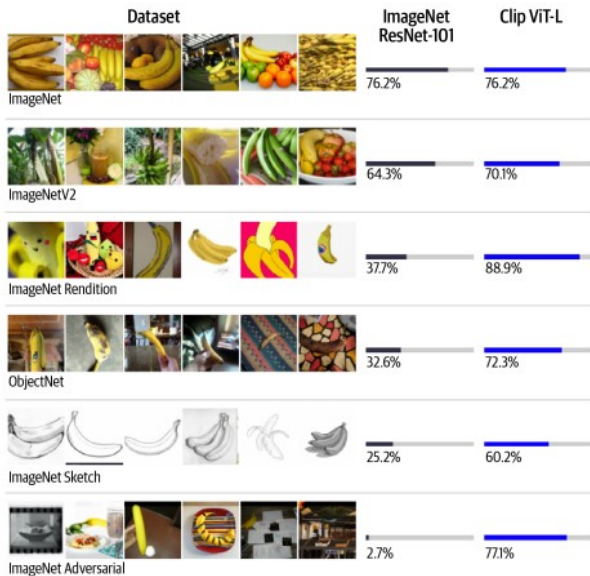


Figure: CLIP performs well on a wide range of image labeling datasets

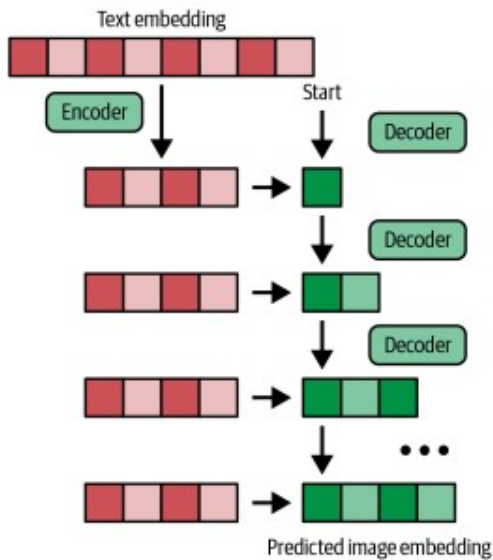


Figure: A simplified diagram of the autoregressive prior of DALL.E 2

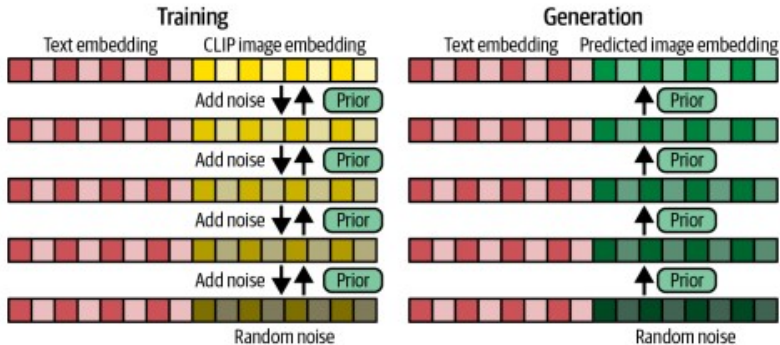


Figure: A simplified diagram of the diffusion prior training and generation process of DALL.E 2

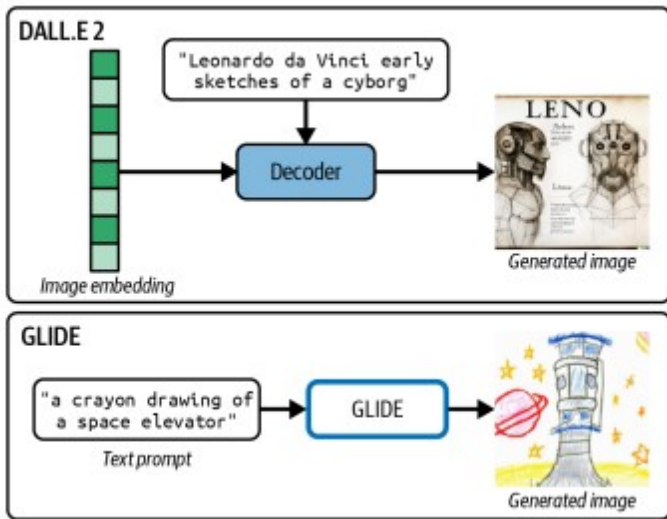


Figure: A comparison between DALL.E 2 and GLIDE—GLIDE trains the entire generative model from scratch, whereas DALL.E 2 makes use of CLIP embeddings to carry information forward from the initial text prompt

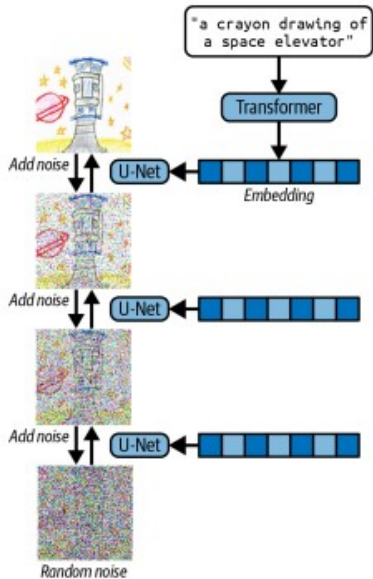


Figure: The GLIDE diffusion process

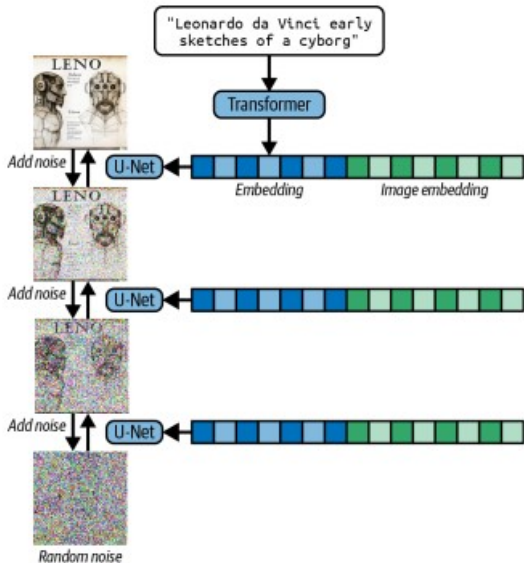


Figure: The DALL.E 2 decoder additionally conditions on the image embedding produced by the prior

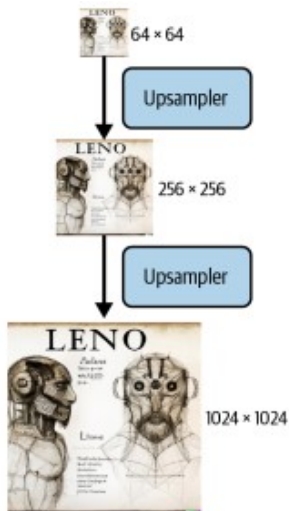


Figure: The first Upsampler diffusion model converts the image from 64×64 pixels to 256×256 pixels while the second converts from 256×256 pixels to $1,024 \times 1,024$ pixels

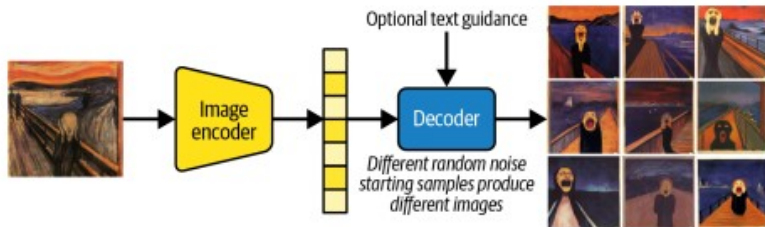


Figure: DALL.E 2 can be used for generating variations of a given image

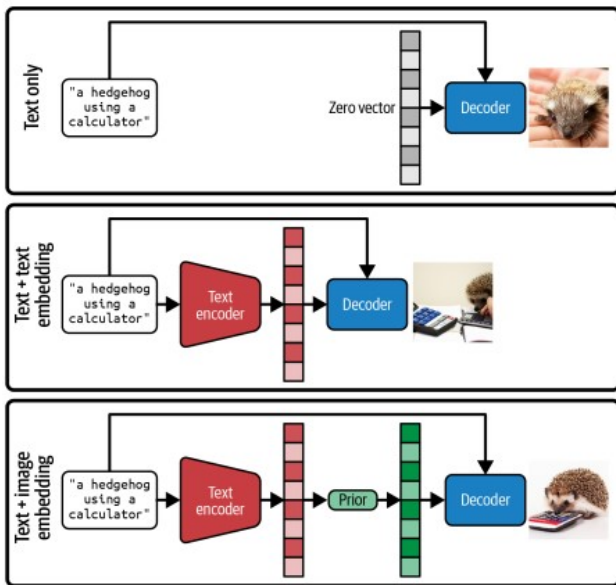


Figure: The prior provides the model with additional context and helps the decoder to produce more accurate generations

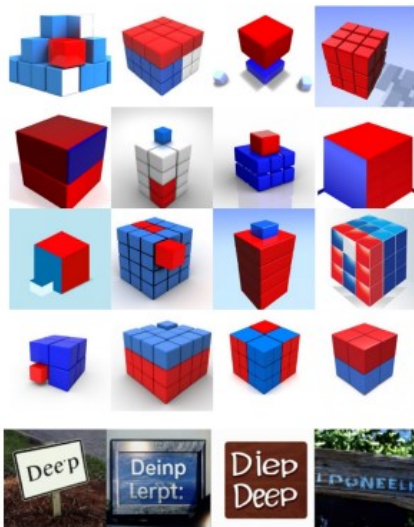


Figure: Two limitations of DALL.E 2 lie in its ability to bind attributes to objects and reproduce textual information—top prompt: “A red cube on top of a blue cube”; bottom prompt: “A sign that says deep learning”

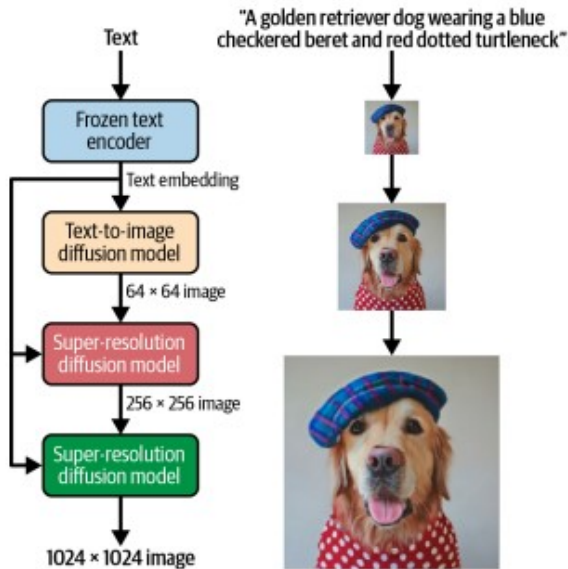


Figure: The Imagen architecture

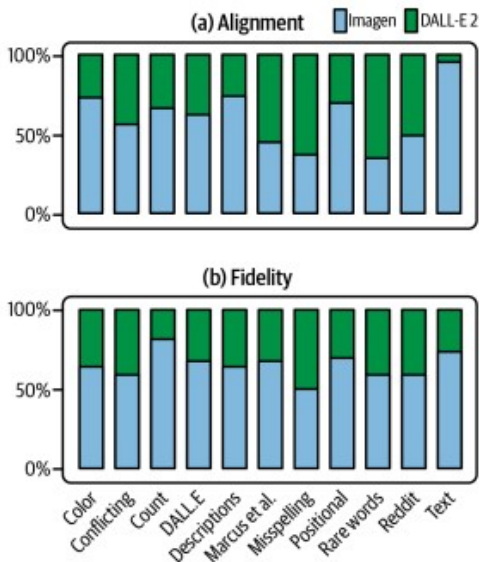


Figure: Comparison of Imagen and DALL.E 2 on DrawBench across alignment and image fidelity



Three spheres made of glass falling into the ocean. Water is splashing. Sun is setting.



Vines in the shape of text "Imagen" with flowers and butterflies bursting out of an old TV.



A strawberry splashing in the coffee in a mug under the starry sky.

Figure: Example Imagen generations

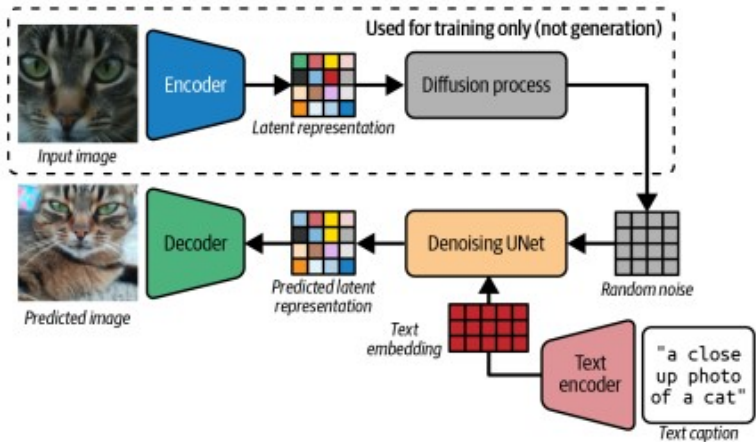


Figure: The Stable Diffusion architecture



"an insect robot preparing a delicious meal"



"a high tech solarpunk utopia in the the Amazon rainforest"



"a small cabin on top of a snowy mountain in the style of Disney, artstation"

Figure: Example outputs from Stable Diffusion 2.1

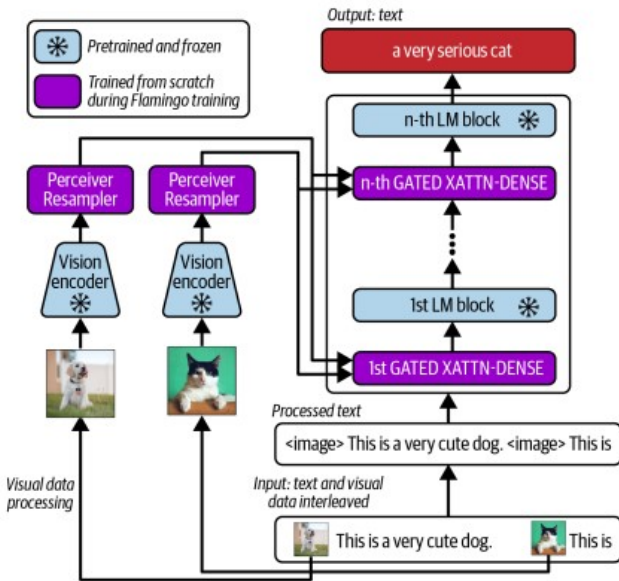


Figure: The Flamingo architecture

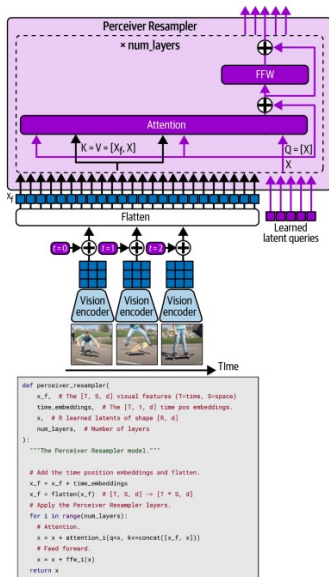


Figure: The Perceiver Resampler applied to video input

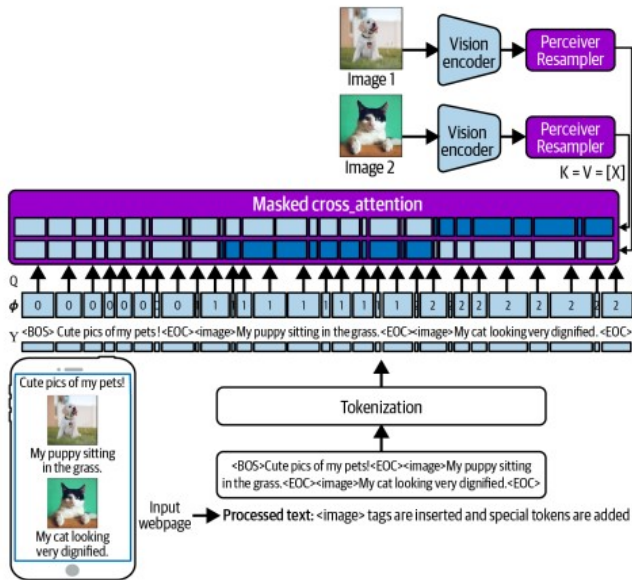


Figure: Masked cross-attention (XATTN), combining vision and text data—light blue entries are masked and dark blue entries are nonmasked

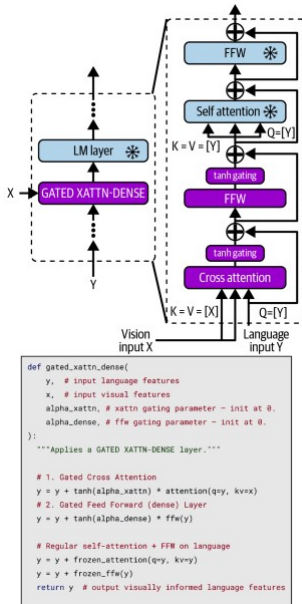


Figure: A Flamingo Language Model block, comprising a frozen language model layer from Chinchilla and a GATED XATTN-DENSE layer

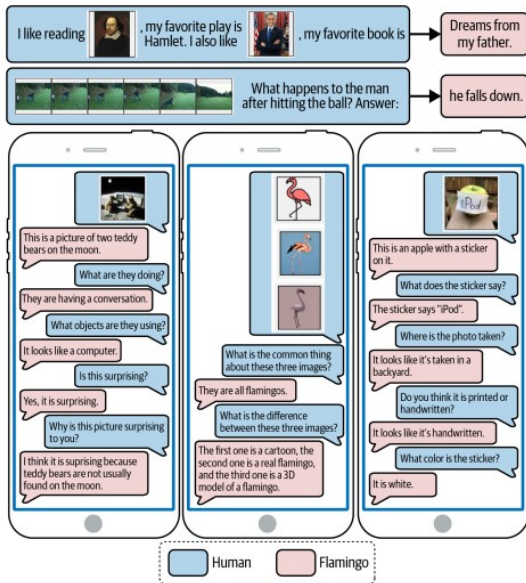


Figure: Examples of inputs and outputs obtained from the 80B parameter Flamingo model